

Software Engineering



Dr. Fatma Eskander
Math.department, Faculty of Science, Mansoura
University

Lecture 4

Use-Case & Activity Diagrams



Part 1: **Foundational Concepts**

Business Processes vs. Business Rules

- ❑ Before we model a system, we must understand the business. Two key concepts are often confused but are fundamentally different.
- ❑ **A- Business Process: The "How "**
- ❑ **Definition:** A collection of linked, structured activities or tasks to achieve a specific goal.
- ❑ **Characteristics:**
 - Has a **start** and an **end**.

Business Processes vs. Business Rules

- Involves multiple **steps** or **activities**.
- Often crosses departmental boundaries.
- Can be visualized as a **flowchart**.

□ Examples:

"Process a Customer Order"

"Hire a New Employee"

"Handle a Customer Support Ticket"

Business Processes vs. Business Rules

- ❑ **Visual:** A simple flowchart icon.

```
[Start] -> [Receive Order] -> [Check Stock] -> [Process Payment] -> [Ship Order] -> [End]
```

- ❑ **B- Business Rule (The "What/Why")"**
- ❑ **Definition:** A specific, testable directive that is under business control and serves to assert business structure or influence the behavior of the business.

Business Processes vs. Business Rules

❑ Characteristics:

- Defines **criteria** for decision-making.
- Often static and does not describe a sequence.
- Enforces business policies. تنفيذ سياسات الأعمال

❑ Examples (related to the "Process Order" example):

- "A customer must have a valid email address."
- "Orders over \$100 qualify for free shipping."
- "Payment must be approved before an order can be shipped."
- "Only a manager can approve a refund over \$50."

Business Processes vs. Business Rules

❑ Key Difference: An Analogy

❑ Making a Cup of Tea

Business Process (The "How"):

1. Boil water.
2. Place tea bag in cup.
3. Pour boiling water into cup.
4. Let it steep for 3 minutes.
5. Remove tea bag.

Business Processes vs. Business Rules

❑ Key Difference: An Analogy

Business Rules (The "What/Why"):

1. The water must be at 100°C to be considered "boiling."
2. The steeping time must be *at least* 3 minutes.
3. The cup must be clean.

❑ Conclusion:

The **Process** is the sequence. The **Rules** are the constraints and definitions that govern the steps within that sequence



Part 2:

Use-Case Diagrams

Modeling User Goals & System Functionality

□ Use-case diagram

What is it?

A use-case diagram is a high-level, visual representation of a system's **functional requirements** as perceived by **external users** (actors).

Primary Purpose:

- To capture the **core functionalities** of a system.
- To identify **who** (actors) interacts with the system and **what** (use cases) they need to do.
- It answers the question: "**What should the system do for each user?**"

Core Components of a Use-Case Diagram

1. System Boundary:

- ❑ A box that defines the scope of the system being modeled.
- ❑ Use cases live inside; actors live outside.

2. Actor:

- ❑ A role played by an external entity (human, another system) that interacts with the system.
- ❑ **Example:** Customer, Manager

Core Components of a Use-Case Diagram

3. Use Case:

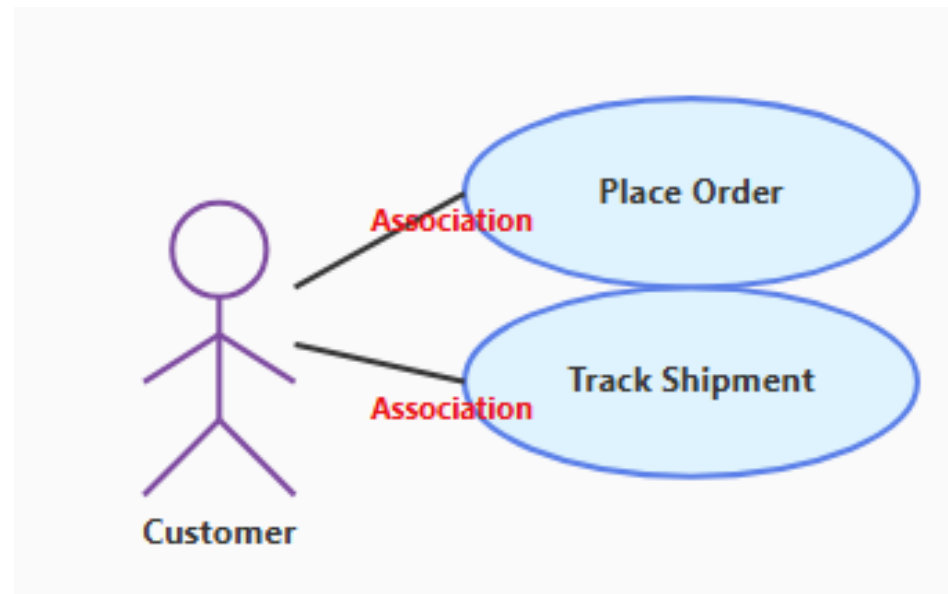
- ❑ An oval representing a specific unit of functionality that the system provides for the actors.
- ❑ It's a **goal** of an actor.
- ❑ **Example:** Place Order, Track Shipment, Generate Report

4. Relationships:

- ❑ **Association:** A simple line connecting an actor to a use case, representing communication or interaction between them.

Core Components of a Use-Case Diagram

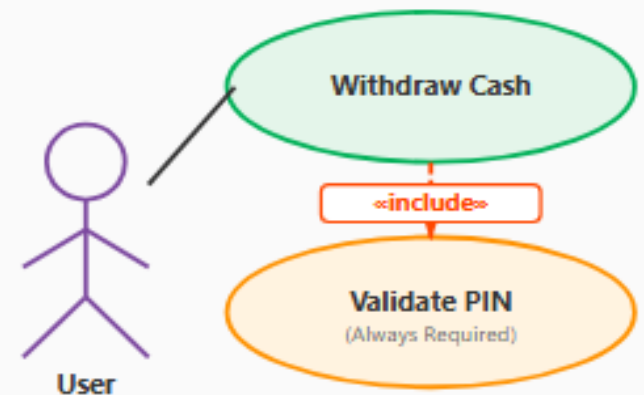
- ❑ **Meaning:** The actor participates in or initiates the use case.



- The **Customer** (actor) initiates and participates in the **Place Order** use case
- The **Customer** (actor) also initiates and participates in the **Track Shipment** use case
- The solid lines indicate that the customer has **direct access** to these system functions

Core Components of a Use-Case Diagram

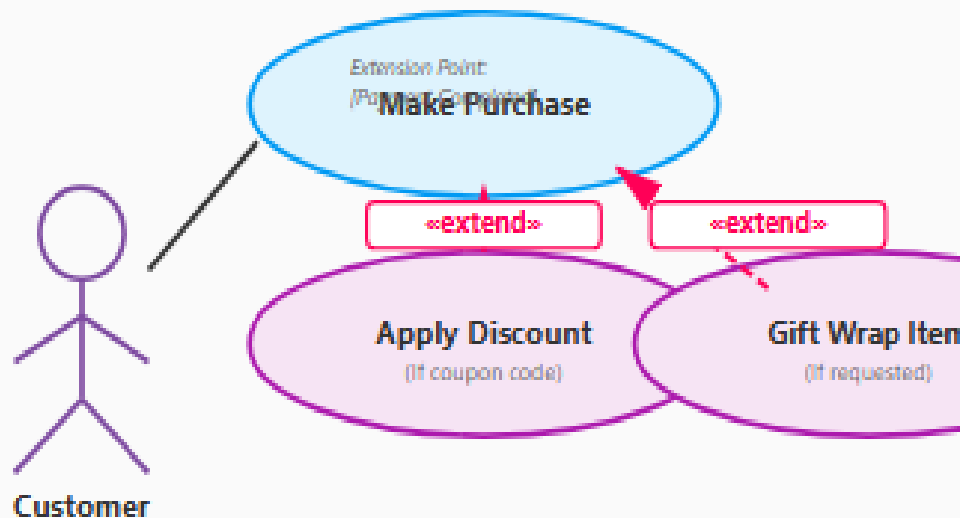
- **Include:** A mandatory relationship علاقة إلزامية. "Use case A *must* include the functionality of use case B.



- **Withdraw Cash** is the base use case (what the user wants to do)
- **Validate PIN** is the included use case (required step that must happen)
- The dashed arrow points FROM "Withdraw Cash" TO "Validate PIN" showing the dependency
- Every time a user wants to withdraw cash, the system **MUST** validate the PIN first
- There is **NO scenario** where you can withdraw cash without PIN validation

Core Components of a Use-Case Diagram

- ❑ **Extend:** An optional dependency where one use case may extend the behavior of another use case under certain conditions.
- ❑ **Meaning:** "Use case B **MAY** extend use case A under specific conditions." It's conditional.



Core Components of a Use-Case Diagram

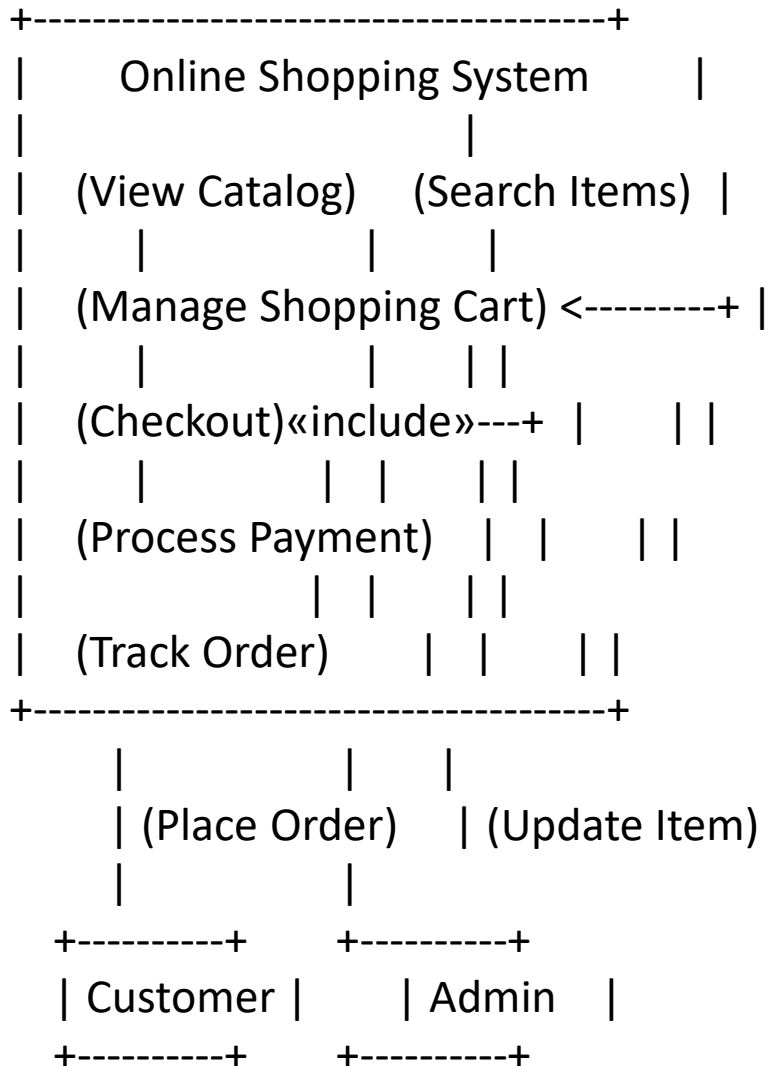
- **Make Purchase** is the base use case (the main functionality that works independently)
- **Apply Discount** and **Gift Wrap Item** are extension use cases (optional add-ons)
- The dashed arrows point FROM the extensions TO "Make Purchase" showing what they enhance
- **Apply Discount** only happens IF the customer enters a valid coupon code
- **Gift Wrap Item** only happens IF the customer selects the gift wrap option
- The purchase can complete **successfully WITHOUT** these extensions

Core Components of a Use-Case Diagram

Quick Comparison

Aspect	Association	Include «include»	Extend «extend»
Line Type	Solid line	Dashed arrow	Dashed arrow
Direction	No specific direction	Base → Included	Extension → Base
Between	Actor ↔ Use Case	Use Case → Use Case	Use Case → Use Case
Execution	Direct interaction	MANDATORY - Always executes	OPTIONAL - Conditional execution
Purpose	Show who uses the system	Reuse common functionality	Add optional behavior

EX



□ **Actors:**

□ Customer, Admin

□ **Use Cases:**

□ View Catalog, Checkout, Place Order

□ **Relationship:**

□ Checkout *must* include Process Payment



Part 3:

Activity Diagrams

Introduction to Activity Diagrams

❑ Activity diagram

What is it?

An activity diagram is used to model the **dynamic aspects** of a system. It represents the **flow from one activity to another**.

Primary Purpose:

- ❑ To model the **detailed workflow** of a business process or a complex use case.
- ❑ To show the **parallel, and sequential flow** of activities.
- ❑ It answers the question: "How is this process or functionality accomplished **step-by-step**?"

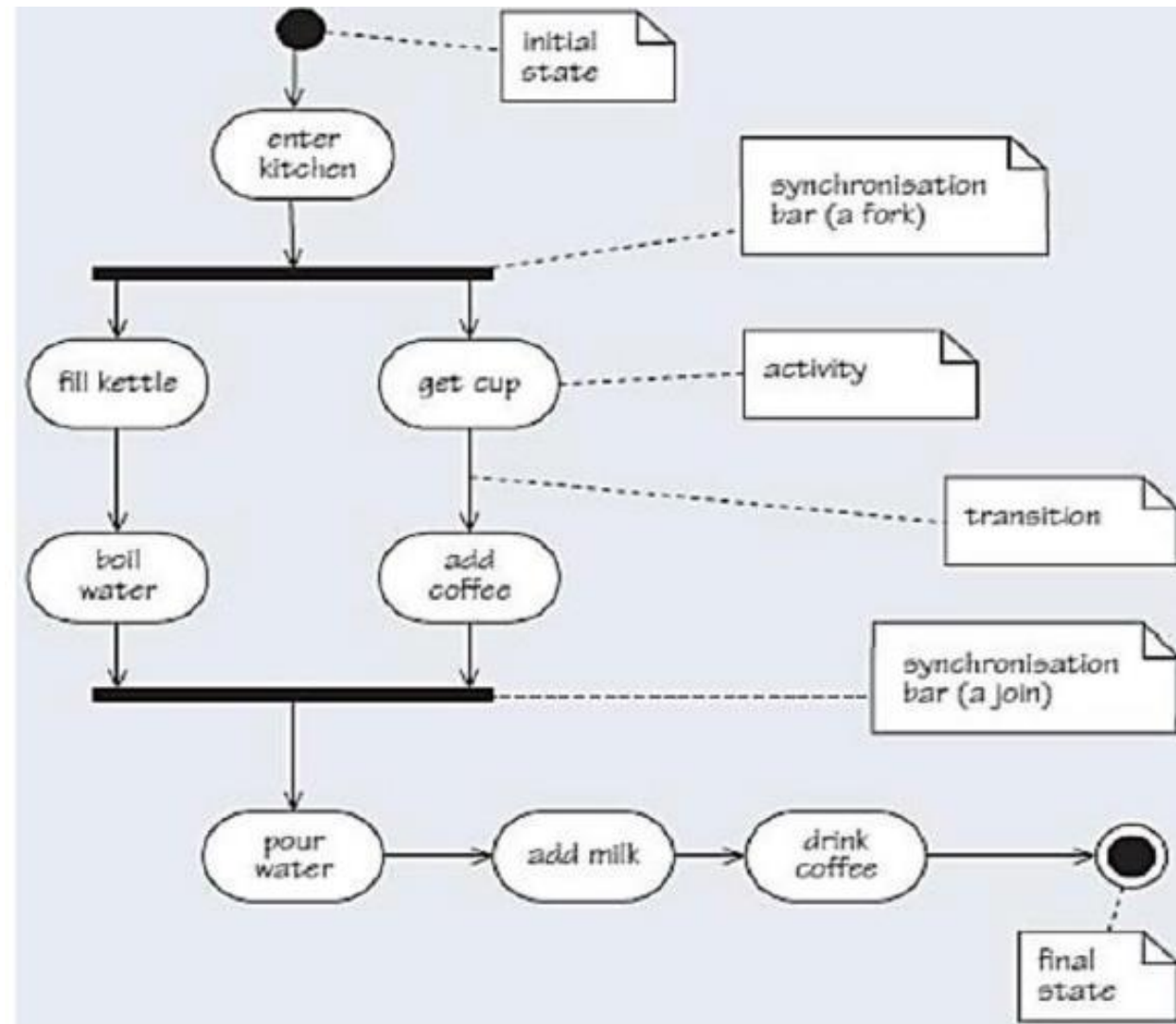
Core Components of an Activity Diagram

- **Initial Node:** (Solid Circle) The start of the workflow.
- **Activity/Final Node:** (Circled Solid Circle) The end of the workflow.
- **Action/Activity:** (Rounded Rectangle) A single step in the process.
- **Control Flow:** (Arrow) Shows the sequence of execution.
- **Decision Node:** (Diamond) A branch in the flow (often with [guard conditions]).
- **Merge Node:** (Diamond) Brings multiple flows back together.
- **Fork/Join Node:** (Solid Bar) Represents the start/end of parallel activities.
- **Swimlanes:** (Columns) Partition the diagram to show which actor or department is responsible for which activities.

Core Components of an Activity Diagram

Element	Meaning	Symbol
Activities	Business processes and tasks	
Start node	Start of process	
End node	End of process	
Transition	move from one activity to another	
Synchronization bar	represents fork and join . Where fork represents the concurrent activities, and join means end of concurrent activities and is used to move to next sequence activity after finishing concurrent ones.	
Decision node	represent alternative ways out of an activity. It has 2 outgoing arcs	
Merge node	brings together alternative flow. It has one outgoing flow.	
Guard	represents the condition (Boolean test); Guards are associated with transitions. A transition takes place only if its guard evaluates to true at the time the flow of control reaches that transition.	[<i>condition</i>]
Swimlane	Swimlanes group activities associated with different roles . Each swimlane shows who is responsible for which set of related activities.	

Ex: Activity Diagram Example:



Use-Case Diagram vs. Activity Diagram

Feature	Use-Case Diagram	Activity Diagram
Level	High-Level (What)	Low-Level (How)
Focus	Actors & Goals	Flow of Activities
Scope	System Scope & User Interactions	A single Process or Use Case
Answer	"What does the system do?"	"How is a specific task done?"